

Accountability over transparency

A heuristic path to calibrated trust in LLM medical advice

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Abbreviations: AI = artificial intelligence, LLM = large language model.

When a 37-year-old man who suffered from a persistent sore throat was told by ChatGPT that cancer was extremely unlikely, he delayed seeking medical care until the stage-4 esophageal adenocarcinoma was diagnosed.^[1] In another case, a 60-year-old man replaced dietary sodium chloride with sodium bromide after consulting ChatGPT for medical advice and later developed bromism with neuropsychiatric symptoms.^[2] Critically, such incidents are not isolated but globally present. Recent evidence suggests that users, especially nonexpert users, place substantial trust in large language model (LLM) medical advice regardless of content accuracy.^[3] This misalignment between content reliability and user expectation is commonly described as trust miscalibration.^[4] In response to such concerns, regulations such as the European Union's AI (artificial intelligence) Act and China's Interim Measures for Generative AI Services have been increasingly emphasizing AI transparency as the primary remedy. The underlying assumption is that if users are explicitly informed that they are interacting with AI (i.e., subject transparency) with limitations and uncertainty (i.e., technology transparency), they will adjust their reliance accordingly.

However, transparency is necessary but not sufficient, particularly for nonexpert users seeking LLM medical advice.^[5] The Heuristic-Systematic Model provides a useful lens for illustration: systematic processing involves effortful, analytic evaluation of arguments, whereas heuristic processing relies on low-effort cues and simple decision rules.^[6] For nonexpert users, systematic processing is often unrealistic, as they lack the clinical knowledge needed to assess medical claims, may be pressed for time or anxious, and are therefore likely to default to heuristic processing. Even when transparency cues are heuristically accessible (e.g., "I am an AI" or performance statistics), they only inform users what the system is and how accurate it might be, and do not respond to the question of "Who is answerable if the advice causes harm?" Critically, this informational gap can turn transparency into a source of overreliance rather than caution. Moreover, prior study

has already shown that AI explanations can increase users' acceptance of recommendations irrespective of their correctness,^[7] and leading voices have warned of the "false hope" that explainability alone can make AI safe for clinical use.^[5] Thus, for users, especially those who are nonexperts, transparency, despite its intuitive appeal, falls short as a trust-calibration mechanism precisely.

1. Accountability as a heuristic cue for trust calibration

1.1. Defining accountability from a user-centric perspective

From a user-centric standpoint, the clinically critical question is not only "How accurate is this system?" but also "Who is responsible if this advice harms me?" Therefore, accountability can be understood as socially grounded signals of responsibility, recourse, and consequence. It refers to cues that indicate who bears the consequence of errors, what avenues for complaint or redress exist, and whether institutional oversight is in place, all of which can be processed rapidly and intuitively without clinical expertise.

1.2. Contrasting accountability cues with transparency cues

We acknowledge that both accountability cues and transparency cues are simple, salient, and heuristically accessible. Yet they differ fundamentally in their social contents. Transparency cues are purely descriptive: they inform users what the system is and what its limitations are, but they do not respond to the relational question of answerability. Accountability cues, by contrast, speak directly to that relational gap. Naming a responsible entity, for example, tells the users that the output is not a disembodied probabilistic statement, but a claim institutionally backed by an organization subject to reputational and legal pressure. The difference means that while transparency cues inform, accountability cues anchor the interaction in a social world, activating existing schemas of professional reputation, legal liability, and consumer protection.

1.3. Psychological mechanism: "Someone is responsible; errors therefore carry consequences"

The cognitive effect of accountability cues can be summarized as a rapid, heuristic inference: someone is responsible; errors, therefore, carry consequences. This line of reasoning is consistent with the concept of "structural assurance" in digital trust research, which refers to the belief that institutional safeguards (e.g., guarantees, regulations, legal recourse, or oversight procedures) are in place to protect users. For users, especially those who are nonexperts, such safeguard does not require them to systematically

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evaluate the medical soundness of the advice when seeking LLM medical advice. Instead, it could prompt a protective and cautious stance in which users treat LLM output as provisional, constrained, and potentially challengeable. In high-stakes medical contexts, this distinction can be decisive: it may lead users, especially nonexpert users, to seek confirmation from professionals rather than act on potentially hazardous LLM output without verification. In this sense, calibrated trust is achieved through a social inference that modulates reliance automatically.

2. Proposed accountability cues for LLM health advice interfaces

For practical considerations, we suggest 3 accountability cues that could be incorporated into LLM health advice interfaces.

2.1. Cue 1: Name the accountable entity

A simple statement such as “Answers provided by [Company X]” or “Answers reviewed by [Hospital Y’s clinical AI team]” immediately tells users who, if anyone, is responsible. Explicitly naming the accountable entity serves as a critical reputational signal,^[8] transforming LLM output into a socially grounded claim tied to institutional authority. This allows users to infer whether the advice is institutionally backed.

2.2. Cue 2: Provide a visible redress channel

A prominently placed button labeled “Report harmful advice” signals that users have a route for recourse. Relevant research indicates that institutional safeguards exist to support trust in digital environments.^[9] This cue, therefore, allows users to challenge LLM output, signaling that recourse exists, that someone is watching, and that the system is not infallible.

2.3. Cue 3: Specify accountable task boundaries in plain language

Generic warnings (e.g., “LLMs have limitations”) are easy to ignore. A more useful cue would be specific and written in plain language, such as “This LLM was tested on 1000 common symptoms with 85% accuracy. It has not been tested on rare diseases or pediatric cases. Please do not rely on the answer without clinician confirmation.” Drawing on the concept of “Model Card,”^[10] this cue helps users assess whether the LLM capabilities fit their specific medical task. Otherwise, they may be more likely to seek professional medical care.

3. Tailoring accountability to user profiles and clinical stakes

While the 3 proposed accountability cues offer a general design framework, their effectiveness depends on how well they are matched to the diversity of users and the risk of clinical task. Accountability design cannot be one-size-fits-all. Even for non-expert users, they are not a homogeneous group, and their ability to interpret accountability cues varies systematically with health literacy, emotional state, and digital experience. For users with lower health literacy, a plain label such as “No doctor reviewed this answer” can serve as an actionable signal. For anxious users, a visible cue such as “Report harmful advice” may be easier to notice than a lengthy disclaimer. For digitally literate users accustomed to using AI, a boundary statement such as “Not tested on rare disease datasets” can signal that this LLM medical advice requires a different level of caution.

The clinical stakes of the advice being sought also matter. Three levels can be distinguished from low-stakes health education to high-stakes action-oriented decisions about treatment. For health education tasks, such as “What is influenza?,” an accountable entity or a general boundary statement may be sufficient. For self-triage questions, such as “Should I go to the emergency department for chest pain?,” users usually benefit from clearer accountability cues, such as “Please seek urgent care for severe or worsening symptoms.” For treatment scenarios, such as “Can I stop this drug?” or “Can I replace this medication?,” the interface should explicitly state whether the advice has been clinically reviewed and whether the LLMs have been evaluated for this case.

4. Conclusion

The growing use of LLMs for medical advice exposes the issue of trust miscalibration. As we have argued, users, particularly non-expert users, predominantly process health information heuristically. Transparency cues, however well-intentioned, lack the social-institutional meaning that heuristic processing requires to generate caution. Accountability cues fill this gap: naming the accountable entity gives the output a reputational anchor; a visible redress channel reminds users that the system is not infallible; and accountable task boundaries in plain language turn abstract limitations into protective, actionable judgment aids. Accountability is not a complete solution, and transparency remains indispensable. But the path toward safer and more responsible LLM use in medical contexts runs through designing for how users, particularly nonexperts, actually process information heuristically and socially, and that design must be anchored in accountability.

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Ethical Statement

Not applicable.

Conflict of Interest

The authors have no conflicts of interest to disclose.

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Data Availability Statement

All data generated or analyzed during this study are included in this published article [and its supplementary information files].

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AI Statement

During the preparation of this work, the authors used ChatGPT and DeepSeek to assist with language polishing. After using these tools, the authors reviewed and edited the contents as needed and take full responsibility for the content of the publication.

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